



Adapting Higher Education in the AI Era: Global and Pakistani Perspectives

Introduction

The rapid rise of **AI tools in education** – from text generators like ChatGPT to code assistants like GitHub Copilot – is forcing higher education to critically reevaluate traditional practices. Students now have at their fingertips AI systems that can instantly produce essays, solve problem sets, or generate programming code. This unprecedented access raises urgent questions: Are our assessments still valid measures of learning when AI can outperform humans at rote tasks? How can we ensure originality and integrity in student research when generative AI can ghostwrite content? More fundamentally, **what skills should universities cultivate** in an age where knowledge recall and routine coding can be automated? These questions are being grappled with globally, including in Pakistan, where regulators like the Higher Education Commission (HEC) are formulating policies on ethical AI use ¹ ² .

This report presents a concrete, critical analysis of how higher education should evolve in response to the widespread use of AI by students. It examines tangible reforms in **assessment design, research thesis evaluation standards, and pedagogical practices**. It also offers **infrastructure and policy recommendations** for universities – both globally and in Pakistan – to promote academic integrity, deep learning, and fair evaluation in an AI-pervasive environment. Example initiatives and emerging best practices from around the world are cited to ground these proposals.

Rethinking Assessment Design in an AI-Driven World

Traditional assessment formats – closed-book exams, recall-based quizzes, formulaic coding assignments – are increasingly ill-suited to an era in which AI systems excel at factual recall and boilerplate problem-solving. As one commentator observed, *“AI performs best at what universities have long assessed — memorization and recall. If machines can do it better, perhaps our task isn’t to compete, but to evolve.”* ³ In other words, AI’s capabilities should prompt a shift in **what** and **how** we test for student “mastery.”

Open-Book vs. Closed-Book Exams: Early reactions to AI in academia fell along a spectrum. Some institutions doubled down on traditional proctored exams to *wall off* AI; for example, **Stanford Medical School** found ChatGPT outscoring their students on open-book clinical exams and swiftly reverted to closed-book, in-person tests ⁴ . Australia’s Group of Eight universities similarly announced in 2023 a return to supervised **“pen-and-paper” exams**, explicitly barring AI usage ⁵ . These moves can curb AI-assisted cheating by forcing students to rely on memory under surveillance. However, they carry significant downsides. Stanford’s faculty noted that while closed-book tests restored independent reasoning, they *“no longer assess students’ abilities to gather information from sources – a crucial skill in clinical care.”* ⁴ In essence, a blanket ban on AI during exams risks **testing the wrong things** – privileging rote recall over the ability to find and evaluate information. Forward-looking educators argue that mastery in the AI age means knowing **how to use AI and other resources effectively** rather than pretending they don’t exist ⁶ ⁷ . A balanced approach is needed: certain high-stakes exams (e.g. professional licensing tests) may remain

closed-book for now, but many assessments should be redesigned to *embrace* controlled use of tools and focus on higher-order thinking.

Authentic and Higher-Order Assessments: To ensure academic integrity while fostering real learning, universities are shifting toward assessment formats where AI offers less of an advantage. Research and early pilots highlight several effective strategies:

- **Practical, Real-World Tasks:** Designing exam questions and assignments that require applying knowledge to novel, complex scenarios can reduce students' ability to rely on AI. Such tasks demand context-specific judgment and creativity that generic AI often cannot easily replicate ⁸. For example, instead of asking students to **recall** definitions or write basic code (which AI can do), an assessment might present a real-world case study or data set and require the student to analyze it, make recommendations, or design a solution. These authentic challenges force learners to synthesize concepts in ways that go beyond regurgitation ⁸.
- **Open-Ended Projects and Presentations:** **Project-based learning** has emerged as one of the most recommended assessment strategies in the AI era ⁹. Students might be tasked with a capstone project, research investigation, or design task that spans weeks, with periodic checkpoints. This dilutes the impact of any single AI-generated response and emphasizes sustained critical thinking. In a systematic review of AI-era assessment methods, project-based assessments were the most frequently cited innovation, followed by peer assessments and oral exams ⁹ ¹⁰. **Oral presentations or defenses** add another layer of integrity: students must explain and justify their work in their own words, making it harder to hide behind AI-written content. Notably, the Australian National University responded to ChatGPT by increasing the use of "*laboratory activities and fieldwork, timed exams and more oral presentations*" in lieu of take-home essays ¹¹.
- **Oral Exams and Viva Voce:** Oral examinations (vivas) and interactive assessments are regaining prominence in the AI age. Because generative AI cannot impersonate a student in real-time dialogue, oral questioning can effectively verify a student's own understanding. Examiners can probe with follow-ups, ask for clarification or justification of answers, and detect if a student is merely reciting superficially learned content. Some universities are training faculty to incorporate **viva-style follow-ups** even in written exams – e.g., having students briefly explain how they arrived at an answer, either in writing or via a short interview, to ensure it reflects their thought process. In computing courses, instructors increasingly use **code walkthroughs** (asking students to walk through their code logic verbally) or **live coding sessions** where students must write or debug code on the spot ¹². Such process-oriented evaluations make it far more difficult to submit AI-generated work that one doesn't understand.
- **Portfolio and Stepwise Assessments:** Another approach is to assess the *process* of learning in addition to the final product. In practice, this means requiring students to submit intermediate drafts, design sketches, or code development logs. By evaluating how a student's work evolved (with reflections on decisions or iterations), instructors can better tell if the work is the student's own. This reduces opportunities for a last-minute AI solution to replace genuine effort. Some computer science programs now ask students to turn in **version histories** or use version control for coding assignments, so that the progression of code (and any unusual jumps that might indicate AI use) is visible ¹³. Similarly, essay assignments might include an in-class brainstorming or outline

submission, to be followed by a final essay – discouraging students from outsourcing the entire task to AI.

- **Higher-Order Questioning:** Where traditional exams are still used, questions are being reoriented toward higher-order cognitive skills. Instead of testing mere recall of facts or theories, questions prompt analysis, evaluation, or creativity. For instance, rather than *“What is the definition of X?”*, a question might pose a scenario and ask, *“How would you apply concept X to solve Y, and why?”* Likewise in coding, rather than *“Write a function to do X,”* an exam might provide a piece of AI-written code and ask the student to *identify bugs or explain and improve* the code. This aligns with the philosophy that assessment should measure *“less about what you know and more about how you think.”* ¹⁴

Crucially, these innovative formats aim to test **skills that AI cannot easily automate**. Studies find that strategies like project work, oral exams, and peer evaluation specifically target **critical thinking and creativity**, which remain distinct human advantages over current AI ¹⁰. By engaging students in active problem-solving and requiring original contributions, such assessments make cheating with AI more difficult and less rewarding. They also better prepare students for real-world challenges. Indeed, educators argue that the world *“no longer cares how much students know. What matters far more is what they can do with what they know.”* ¹⁵ In a workplace where AI is ubiquitous, employees will be valued not for memorizing facts, but for applying knowledge creatively, collaborating, and exercising sound judgment – competencies that these reformed assessments cultivate.

The Fate of Memorization-Based Testing: One of the most pointed debates is over the role of **memorization** in curricula and exams. The availability of AI and the internet calls into question the longstanding exam practice of testing rote memory under closed-book conditions. On one hand, foundational knowledge still matters – doctors must know medical fundamentals, and engineers the laws of physics, without constantly consulting a device. Additionally, excessive dependence on AI for basic facts could leave students with *“no more than certificates of attendance”* rather than real competencies ¹⁵. On the other hand, if an AI can instantly retrieve a formula or historical date, spending valuable instructional time and stress on memorizing such details may be misguided. Education thought leaders note that universities have historically over-emphasized information retention at the expense of higher-order skills, a gap now laid bare by AI ¹⁵. The consensus emerging is that **we must recalibrate what we expect students to commit to memory**. Core conceptual understanding and mental frameworks are still essential – they enable students to contextualize and vet AI-provided answers – but minute trivia and boilerplate methods are less crucial to test in isolation. Instead of eliminating memory work entirely, educators should focus on **meaningful knowledge**: students should internalize key principles and schemas of a domain, which then empower them to ask the right questions of AI and detect when its output is off-base. In sum, memorization should become a means to an end (deep comprehension and application), not an end in itself. As we will discuss, this shift has implications for how we train students (and what we value as “mastery”) in the age of AI.

Upholding Research Integrity: Evolving Thesis and Dissertation Standards

At the MPhil and PhD levels, the stakes of AI disruption are especially high. Advanced researchers are expected to produce *original* scholarship – formulating new ideas, generating novel data or analysis, and writing at length – all of which generative AI can *simulate* with varying degrees of success. The question is

how to ensure that a thesis truly reflects a scholar's own contribution and understanding, not the handiwork of an algorithm. This necessitates updated standards for originality, explicit **AI assistance disclosure**, and better preparation for examiners who must evaluate work in this new context.

Originality and Use of AI: The bedrock of doctoral work is that it contributes new knowledge. Universities are therefore drawing a hard line that **outsourcing thinking or writing to AI violates academic integrity**. For example, Pakistan's HEC in 2023 convened an experts committee to draft a *Framework on the Use of Generative AI in Higher Education*. The draft policy makes clear that using AI to produce substantial portions of assignments or theses is considered a form of plagiarism or academic misconduct ¹⁶ ¹⁷. The rationale is that AI-generated text is derived from existing works and often includes uncited, inaccurate, or fabricated information ¹⁶. If a student simply submits such AI output as their own writing, it both misrepresents authorship and risks propagating errors. Thus, HEC's guidance warns that *"any assignment or thesis that is found to have used unauthorized GenAI tools may receive a zero and/or be reported for academic misconduct."* ¹⁷ This strict stance is echoed globally – major academic journals like *Science* have outright banned AI-generated text in submissions and treat its use as scientific misconduct ¹⁸ ¹⁹.

At the same time, outright prohibition of AI use in research writing may be unrealistic and arguably counterproductive. **Guidelines vs. Bans:** Many institutions now allow limited, transparent use of AI as an assistive tool. For instance, **King's College London (KCL)** explicitly permits doctoral students to use generative AI in an *"assistive role"* during thesis writing, such as for improving clarity of English or brainstorming ²⁰ ²¹. The crucial caveat is that this use must remain within certain bounds and be fully disclosed. KCL stresses that students *"should declare their use of generative AI tools and take full responsibility for the content of their submitted thesis,"* and that they *"will not be penalised" for AI use if it is consistent with the guidance and properly declared* ²². Similarly, HEC's draft framework does not ban AI entirely; it acknowledges AI can be used to enhance language and readability (e.g. editing grammar), but **AI is not recognized as an author** and any content contributions from AI must be minimal, carefully verified, and cited ²³. In fact, HEC advises that no more than 5% of a thesis's text should be AI-generated, aligning with updated plagiarism-check thresholds ²⁴ ²⁵. Both KCL and HEC require a formal **declaration of AI usage** to be included with any thesis or dissertation. For example, HEC's suggested student undertaking states: *"I solemnly declare that the research work presented in my ... thesis is solely my own, with no significant contribution from any other person or Generative AI tools. Small contribution/help wherever taken has been duly acknowledged/cited and the complete thesis has been written by me in accordance with the latest policy on use of Generative AI Tools."* ²⁶. This kind of statement, to be signed by the student and endorsed by the supervisor, creates an official record that the student attests to the work's originality and any AI assistance is appropriately acknowledged.

Training Examiners and Detecting AI Text: The responsibility for upholding standards does not lie with students alone. **Thesis examiners and supervisors** must be equipped to identify possible AI influence and enforce the new norms. Key steps include:

- **Awareness of AI Writing Characteristics:** Examiners should be aware of telltale signs of AI-generated text, such as overly generic phrasing, lack of clear narrative voice, or the presence of subtly incorrect facts or citations (AI often produces *hallucinated* references). However, examiners are cautioned against jumping to conclusions based on style alone. KCL's guidance pointedly advises examiners *"not [to] draw negative conclusions about generative AI use on the basis of assumptions about a student's likely English proficiency,"* and also **prohibits examiners from feeding any part of a thesis into an AI detection tool** or into an AI service for analysis ²⁷. These rules recognize that

current AI detectors yield false positives and that uploading a student's work to a third-party AI (to check if it was written by AI) could violate confidentiality. Instead, examiners are encouraged to rely on traditional methods of evaluation: close reading, academic judgment, and rigorous oral examination.

- **Emphasizing the Oral Defense:** The **viva voce (oral defense)** has taken on heightened importance in the AI age. During a thesis defense, examiners can directly probe the candidate's understanding, asking detailed questions about the work. If a student has used AI to write passages without fully grasping them, this should become apparent under questioning. Universities are updating their training for PhD examiners to ensure they ask pointed questions that test authenticity. For instance, examiners may ask a candidate to explain how a particular argument was developed, justify a methodological choice, or elaborate on how their work fills a gap in the literature ²⁸. As KCL notes, *"candidates may be asked to discuss in detail a specific source they cited, or to concisely articulate how their research makes an original contribution"*, precisely to verify that the intellectual work is truly theirs ²⁹. A student who merely pasted an AI-generated literature review without digesting it will likely falter in such discussion. Thus, robust oral examinations act as a safeguard for research integrity. They also send a clear signal to students: you must **understand and own your thesis** inside-out, AI notwithstanding. In fact, KCL warns that if AI was used beyond acceptable limits, the student is likely to *"find themselves unable to satisfactorily answer such questions"* and will risk failing the defense ³⁰.

- **Advanced Plagiarism Detection:** Universities are also exploring technical tools to aid in catching undisclosed AI use. Turnitin (a leading plagiarism detection service) has introduced an AI-writing detector, and independent tools like GPTZero and others have emerged ³¹ ³². These use machine learning to flag writing that statistically resembles AI output. However, current detection tools are far from foolproof – their accuracy might catch 70-80% of AI text, but can be evaded by edited AI text and often generate false accusations ³³ ³¹. A senior Australian official even admitted that reliably **"detecting AI cheating in university assignments is now 'impossible'"** at scale with available technology, calling for a *complete overhaul* of assessment strategies instead ³⁴ ³⁵. Given this reality, institutions like HEC and others place more weight on preventative policies (declarations, honor codes) and personal evaluation (viva) than on hoping for a perfect AI detector. Nevertheless, examiners and supervisors are being briefed on how to use such tools judiciously – perhaps as a preliminary scan – and how to interpret their findings (e.g. requiring a student to orally explain any sections that score suspiciously).

Disclosure and Ethical Use: A positive development is the normalization of **AI assistance disclosure** in academic writing. Just as researchers disclose using a proofreader or statistical consultant, students should disclose if and how they used AI in producing their thesis. Many universities (Princeton, Harvard, George Mason University, to name a few) have issued guidelines that if AI was used, it must be clearly acknowledged either in a dedicated section or in footnotes ³⁶ ³⁷. HEC's draft policy includes a template for a "Declaration of Generative AI tools in the writing process," where the student would fill in the tool name and purpose (e.g. *"the author used GPT-4 to suggest improvements in grammar and phrasing"*) and affirm that they subsequently reviewed and take responsibility for all content ³⁶. Importantly, **AI cannot be credited as an author** on any research publication or thesis ²³ – reflecting global consensus that authorship implies accountability, which only humans can bear ¹⁹. The disclosure trend has two benefits: it deters covert cheating (since students know they must come clean) and it legitimizes *appropriate* uses of AI,

integrating them as part of the research process. Students learn to see AI as a tool under their control rather than a secret shortcut.

In summary, preserving the integrity of graduate research in the AI era will require a combination of strict **academic honesty rules**, technological tools, and human-centric evaluation. Universities must set clear expectations that theses should largely be the candidate's own work, with any AI assistance minimal and openly acknowledged. Examiners, in turn, should be trained to uphold these standards through keen observation and incisive questioning. The end goal is that a thesis or dissertation remains a credible certification of the student's *original* scholarly capability – not just their ability to prompt an AI.

The Role of Memorization and Manual Skills in an AI Age: Value vs. Obsolescence

One of the most profound pedagogical questions today is what **knowledge and skills** students should still be required to develop *without* AI, and which skills can be de-emphasized now that AI is readily available. Two areas in particular are debated: the value of **memorizing information** and the need to **manually write code or calculations** in an era of intelligent automation.

Memory and Conceptual Understanding: Critics of the old education model argue that it overvalued rote memorization – students cramming facts for closed-book exams – which made sense in a pre-digital age but is unnecessary now. Why memorize dozens of formulas or historical dates when an AI assistant can retrieve them on demand? As one education expert quipped, *“The world simply no longer cares how much students know. What matters far more is what they can do with what they know.”* ¹⁵. This perspective suggests that **knowledge retrieval** is less important than knowledge application. Indeed, AI's prowess at recall has exposed that many traditional tests were essentially measuring skills (memorization, routine procedure) that machines now perform better than humans ³. Persisting in such assessments could be counterproductive – it might encourage students to cheat with AI or to disengage, feeling that their education is out of touch. Moreover, time spent on lower-order drill could instead be spent on higher-order analysis or creativity that adds more value.

However, declaring memorization “obsolete” would be an overstatement. Cognitive science and many educators caution that a certain base of factual and procedural knowledge in long-term memory is *necessary* for critical thinking. A student who has to ask an AI for every basic fact or to solve every simple step may struggle to tackle complex problems, because they have no internal schema to integrate and judge the information the AI provides. For example, a medical student who hasn't memorized symptoms of common conditions might not know if an AI's diagnostic suggestion is plausible. There is also the risk of **AI dependence**: without any memorized knowledge, a learner might believe whatever the AI says, lacking the independent frame of reference to spot errors. As a recent commentary put it, education should not aim to “graduate humans who compete with machines,” but rather **“cultivate independent thinkers”** who can reflect and judge – and that *does* require an internal store of knowledge and experience ³⁸. In short, **memorization is still pedagogically valuable** when it builds foundational understanding and fluency (e.g. knowing vocabulary, key concepts, multiplication tables, syntax of a programming language). What should change is using memorization *as the primary yardstick of learning*. Instead of asking students to regurgitate facts under pressure, we should encourage them to internalize core knowledge and then demonstrate how they can *use* it in analysis or innovation.

Manual Coding and “Show Your Work”: Similar arguments surround **programming and other technical skills**. With AI code generators able to produce working code from a natural language prompt, some wonder if future programmers will need to learn traditional coding at all. Already, tools can complete code or find bugs, raising the question of whether students should struggle with syntax and debugging or simply learn how to effectively prompt an AI. The emerging consensus in computing education is that *learning to code is still essential*, but the approach and emphasis are shifting. A position paper by the Raspberry Pi Foundation (2025) offers several reasons why *“kids still need to learn to code in the age of AI”*. First, even if AI can generate code, society will *“need skilled human programmers who can think critically, solve problems, and make ethical decisions.”* The AI’s speed in writing code is not a substitute for human oversight – AI will happily produce code that is inefficient or even harmful if not guided properly ³⁹. The human programmer’s role will increasingly be to **formulate problems**, evaluate AI outputs, and refine them, which one can’t do effectively without understanding coding fundamentals. Second, the act of coding is tightly linked to computational thinking. *“The hard cognitive work of reading, modifying, writing, explaining, and testing code is precisely how young people develop a deep understanding of programming,”* the paper notes ⁴⁰. In other words, manual practice is how the brain internalizes algorithmic logic. If students only ever use AI as a crutch, they may never develop true programming intuition or the ability to solve new problems from scratch.

That said, computing educators are re-examining **what kinds of coding tasks** students should do manually. Routine or boilerplate coding (the type an AI can do effortlessly) is less important to insist on. Instead, instructors are focusing on design, problem decomposition, and debugging skills. Students might be allowed to use AI to write chunks of code, but then be asked to analyze and improve that code for efficiency or correctness. This approach treats AI as a *“pair programmer”* or learning partner. It also aligns with the idea that the curriculum may shift **“to focus more on understanding, verifying, and improving AI-generated code/examples rather than on writing code from scratch.”** ⁴¹. **Notably, one academic study observed that “traditional programming assignments are no longer a reliable indicator of a student’s understanding; the link between code production and understanding is broken”* when AI can generate the solution** ⁴². **A student could turn in a perfectly working program written largely by ChatGPT, yet not comprehend why the code works. To counter this, many computer science courses have introduced interactive assessments: for example, after a coding assignment, a student might have to meet with the instructor for a code walkthrough interview, explaining each part of their solution and answering tweaks or extensions. If the student authored and understood the code, they should manage this; if they merely solicited it from AI, they will struggle. In other cases, instructors give in-class coding tests (on paper or simple IDEs without internet) to ensure students can produce basic algorithms on their own. Such measures underscore that students must still *learn to code*, even if in practice they will later use AI assistance. The difference is that now educators emphasize learning how to collaborate with AI** (e.g. writing good prompts, checking AI’s output, integrating multiple tools) in addition to mastering the core concepts manually.**

In summary, **requiring students to memorize or to manually work through problems has both pedagogical value and limitations** in the AI era. The value lies in building mental models, intuition, and independent critical thinking ability – one must *know something* in order to effectively use AI or to go beyond AI. The limitation is that we should not overemphasize rote skills to the detriment of higher-order competencies, nor use them as the sole proof of learning. The educational approach should evolve to: - **Teach and test core knowledge and skills in context**, not in isolation. - Leverage AI to handle low-level details (like boilerplate code or factual lookup) while pushing students to spend more effort on interpretation, strategy, and creativity. - Encourage students to develop **“AI literacy”** – understanding what

AI can and cannot do, and learning to verify its outputs against their own knowledge. This way, they remain in control of the problem-solving process. As the saying goes, “Program, or be programmed”⁴³ – if we want students to be agents of technology and not passive recipients, we must still empower them with foundational skills and the discernment to know when and how to rely on AI.

Infrastructure and Policy Reforms for an AI-Augmented Academy

To implement the above shifts effectively, universities and educational authorities need to update their **infrastructure, policies, and quality assurance mechanisms**. Below are key recommendations – with a focus on both global best practices and the particular context of Pakistan – to ensure academic integrity and robust learning outcomes in a world where AI is ubiquitous among students.

1. Clear Institutional Policies on AI Use: Every university should establish a transparent policy defining how AI tools may or may not be used in coursework and research. This prevents confusion and aligns expectations between students and faculty. For example, a policy might state that AI-based grammar and style assistance is permitted (akin to using Grammarly), but generating entire essays or coding assignments via AI is prohibited unless explicitly allowed by the instructor. Many universities now require that **if AI is used, it must be disclosed in the assignment or thesis**^{36 37}. Honor codes and academic integrity charters are being updated to include clauses about misusing AI. Pakistan’s HEC has recognized this need by framing guidelines to “reinforce the autonomy and responsibility of Higher Education Institutions in ensuring the authenticity of research and combating plagiarism with regards to Generative AI tools.”² HEC’s draft policy explicitly mandates universities to treat undeclared AI-generated content as plagiarism and to obtain students’ informed consent about permitted AI use in each course^{17 23}. Likewise, education bodies elsewhere (e.g., Australia’s TEQSA and the UK’s Quality Assurance Agency) have issued advisory notes urging institutions to adapt their academic integrity policies in light of AI^{44 45}.

2. Faculty Training and Support: Faculty are at the frontline of implementing new assessment designs and detecting integrity issues. Universities should provide workshops and resources to help instructors redesign their courses for the AI age. This includes training on creating AI-resilient assignments (as discussed earlier), using AI detection tools carefully, and understanding the pedagogical trade-offs. Educators also need forums to share what’s working or not – for instance, a professor might pilot open-book exams with AI-usage allowed and report the outcomes to colleagues. In Pakistan, where many faculty may not yet be well-versed in AI tools, the HEC and university leadership can organize capacity-building sessions. Topics might cover: *How to use generative AI as a teaching aid*, *how to adjust grading rubrics when students use AI*, and *case studies of academic dishonesty involving AI*. By raising faculty AI-literacy, universities enable teachers not only to police AI misuse but to positively incorporate AI where it enhances learning. A global example is the “ChatGPT Edu” pilot programs at institutions like Duke University and Virginia Tech, which provided select faculty with access to ChatGPT and training to experiment with it in instruction^{46 47}. These pilots treat AI as an opportunity for innovation in pedagogy (e.g., using AI to generate practice problems or to simulate debates in class) rather than solely as a threat.

3. Student Guidance and AI Literacy: On the student side, universities should integrate **AI literacy and ethics** into the curriculum. This means teaching students how generative AI works, its strengths and weaknesses, and how to use it responsibly. Some universities have begun offering seminars on “Effective and Ethical Use of AI in Academic Work.” Students learn, for example, how to prompt AI for studying or brainstorming without violating academic integrity, and how to critically evaluate AI-generated content. The aim is to create a culture where students view AI as a tool to augment their learning (when allowed), not as

a cheating device. In Pakistan, student-facing initiatives could include online tutorials in Urdu/English about the HEC's AI usage guidelines, and mandatory orientation sessions for new university entrants on what constitutes acceptable vs. dishonest use of AI. By demystifying and contextualizing AI, we reduce the temptation for misuse and empower students to use the technology in pedagogically productive ways.

4. Assessment Infrastructure: Universities should invest in **secure assessment environments** and monitoring tools in response to AI. For high-stakes exams, this might entail lockdown browsers or exam software that restricts access to AI tools (as an interim measure when needed). Some institutions use proctoring software enhanced with AI to watch for cheating behaviors during online exams (though these raise privacy concerns and aren't foolproof against AI use on secondary devices). On the flipside, if an instructor *wants* to allow AI use in an exam, the infrastructure should support that too – e.g., providing an “AI-enabled exam browser” that gives access to certain AI tools along with a log of usage for later review. The goal is to have flexibility: teachers can choose the right tool-access level for each assessment. Additionally, as part of assessment redesign, more weight can be shifted to coursework components that are less amenable to cheating (projects, presentations, vivas as mentioned). For those, ensuring adequate **facilities and scheduling** is important – e.g., enough time slots for oral exams, labs equipped for practicals, etc.

5. Quality Assurance in Theses and Research: Education regulators and universities should strengthen the mechanisms for **doctoral and MPhil thesis review** in the face of AI. HEC's approach in its draft policy is illustrative: it requires each thesis to undergo rigorous plagiarism scanning (with updated thresholds for AI text) and for supervisors to personally vouch that the work meets the new criteria ²⁴ ⁴⁸. HEC even contemplates post-award action: if it's later discovered that a thesis was substantially AI-written (hence plagiarized), the university and HEC reserve the right to revoke the degree ⁴⁹. Such strictures send a strong deterrent message. Beyond policies, examiners themselves should be briefed before each thesis defense about the institution's stance on AI. Some universities now include a checkbox or question on the examiner report form like, “Does the candidate's work show evidence of AI tool use beyond what is declared or permitted?” – prompting examiners to comment on it. National bodies (like HEC) can also update **PhD program accreditation standards** to ensure universities have measures for AI. For example, a requirement that each university establish an “AI ethics committee” or assign an officer (perhaps attached to the office of research integrity) to handle cases and queries related to AI in student research. These systemic checkpoints will nudge universities to stay vigilant.

6. Curriculum and Skills Reform: On a broader level, higher education needs to realign curricula to emphasize the human skills that complement AI. Regulatory bodies and accreditation agencies should encourage (or mandate) that programs incorporate **deeper learning outcomes** – critical thinking, problem-solving, creativity, collaboration, and digital literacy. In Pakistan, the HEC can take the lead by updating its **National Curriculum guidelines** for various disciplines to reflect the realities of AI. For instance, the computer science curriculum might add a module on AI-assisted development and its pitfalls. Business and humanities programs could integrate discussions on AI in their fields (e.g., AI in finance, AI in language translation) to make students aware and prepared. Moreover, “alternative assessment models” could be written into policy recommendations – e.g., an HEC circular could suggest that at least X% of assessment in undergraduate courses should be project-based or oral, to reduce reliance on pure written exams. These top-down signals can accelerate the adoption of better practices at the departmental level.

7. Continuous Evolution and Collaboration: Finally, embracing AI in education should be viewed as an ongoing process. Universities should regularly *review and update* their policies as AI tools evolve (the pace of

AI advancement means what is rare today could be common tomorrow). Sharing knowledge is key: what works in a pilot program at one university can be scaled or adapted by others. This calls for **international and national collaboration**. Globally, bodies like UNESCO and OECD are already facilitating discussions on AI's impact on education, producing reports that universities can draw on. Locally, Pakistani universities (and others) could form consortia or working groups to develop unified responses – much like Australia's Group of Eight took a collective stance on redesigning assessment ⁴⁵. HEC's consultation with experts and periodic revision of its AI guidelines (the draft suggests an annual review to keep pace ⁵⁰) is a wise approach. By working together, institutions can stay ahead of the curve rather than each reacting in isolation.

In essence, the infrastructure and policy response must ensure that **academic integrity is maintained without stifling innovation**. Students and faculty should have the tools to leverage AI for learning, under frameworks that keep the playing field fair and the learning outcomes meaningful. The reforms suggested above, from clear guidelines and training to curriculum redesign, aim to create an ecosystem where AI is integrated in a thoughtful, ethical manner. This will help produce graduates who are not only knowledgeable in their fields, but also adept at using emerging technologies responsibly – a critical trait for the future workforce.

Conclusion

The advent of powerful AI tools has been a disruptive force for higher education worldwide. Yet, as with past technological revolutions, it also presents an opportunity to **reimagine and improve** how we teach, learn, and assess. The traditional model of education – characterized by rote learning, high-stakes closed-book exams, and lengthy written theses – is being challenged by machines that can memorize, compute, and even compose text better and faster than students. Rather than seeing this as a threat to be extinguished, educators and policymakers are increasingly viewing it as a catalyst to update an outmoded system. As one analysis put it, *“If machines can do it better, perhaps our task isn't to compete, but to evolve.”* ³

In practical terms, evolving means **shifting our focus to the uniquely human competencies** that define true mastery. Higher education should test not just what students *know*, but what they can *do* – how they think critically, solve novel problems, generate original ideas, and make ethical judgments. AI can be a partner in this process: students can use AI to explore ideas and accelerate routine work, while instructors use it to enrich instruction and personalize feedback. However, the **core learning outcomes and integrity standards must be upheld** by thoughtful assessment design and clear ethical guidelines. The reforms discussed – from real-world assessments and oral defenses to AI usage disclosures and policy overhauls – are all pieces of a new paradigm where learning is deeper and cheating by automation is less of an attractive shortcut.

For Pakistan, in particular, the moment is ripe to leapfrog into modern pedagogical practices. The HEC's proactive efforts in drafting AI policies ², and the dialogue this has opened, should continue. By adopting global best practices and tailoring them to local contexts (e.g. resources constraints, faculty readiness), Pakistani universities can ensure their graduates remain competitive and competent in the AI-augmented global economy. This includes moving away from a culture of **rote memorization and “notes-based” exams** – widely criticized as producing superficial learning – towards one of inquiry, application, and creativity. It also means investing in faculty development and technological infrastructure so that the integrity of qualifications (be it a bachelor's degree or a PhD) is never in doubt, even as students harness new tools.

In conclusion, the widespread use of AI by students is a reality that education systems must face head-on. Clinging to old assessment formats or hoping to catch every AI misuse is a losing battle ^{1 44}. Instead, higher education should adapt by redesigning assessments to outsmart the cheat bots, by evaluating the *process* and *reasoning* not just the final answer, and by instilling in students the mindset to use AI ethically. With well-considered reforms in assessment design, research evaluation, pedagogy, and policy, we can ensure that a degree in the age of AI still stands for genuine learning and intellectual growth. In doing so, universities will fulfill their ultimate mission: producing graduates who are not only knowledgeable, but also agile, critical thinkers – human intellects enhanced, not replaced, by the power of AI.

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